

RUPAM BHATTACHARYA

Senior Engineer | Builds Across Product, Cloud, and AI | M.Sc. AI (TUM) | AWS Pro

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SUMMARY

Senior engineer with 12+ years of experience shipping end-to-end systems across product, infrastructure, and AI. Built and operated platforms serving 25M+ daily users, architected publishing systems used by 200+ brands (Vue / Angular front-ends, Node.js / TypeScript services), and now ships production AI — an agentic developer assistant and a natural-language OpenSearch interface (Next.js / TypeScript + Anthropic API) — used inside the company every day. Equally comfortable writing a Next.js front-end, owning an AWS account structure, evaluating LLMs as deployable artefacts, or growing a team. Looks for problems where the answer requires moving across the stack: product framing, distributed-systems engineering, and applied AI in the same week.

WHAT I BUILD ACROSS

Product Engineering: Internal developer tools as products (customer = engineer); natural-language interfaces to operational data; multi-brand publishing platforms; CMS / content rendering; full lifecycle ownership from discovery to operation

Full-Stack: Next.js, React, Vue, Angular front-ends; Node.js / TypeScript APIs; AWS serverless (Lambda, SQS, SNS, DynamoDB) and event-driven architectures; testing with Storybook, Playwright, Cypress, and supertest in CI

Cloud & Platform: AWS at scale (Solutions Architect – Professional), Terraform / IaC, EKS / ECS Fargate, multi-account architecture, CI/CD (GitLab, GitHub Actions), Datadog observability, cost & vulnerability reduction

Applied AI / ML: Production LLM applications, agentic / tool-calling systems, multi-LLM evaluation pipelines, prompt-versioning & release processes, deep learning (RNNs, CNNs) from M.Sc. AI research

Languages: Python, TypeScript, Node.js, Go, Java, Rust

Practices: Architecture decisions, hiring & mentoring, SLO/SLI, DORA metrics, treating internal platforms as products with measurable adoption

FEATURED WORK

Natural-Language Interface to OpenSearch • ProSieben

- Stateful chat UI (Next.js / TypeScript) where product managers and engineers query the OpenSearch cluster in plain English; Claude translates intent into DSL queries via tool use and generates persistent dashboards — cutting time-to-insight and freeing engineering from ad-hoc reporting requests. Cross-stack: UX, LLM tool design, search engineering, and dashboarding.

GenAI Developer Agent (CLI) • ProSieben

- Agentic CLI orchestrating OpenAI and Anthropic APIs with Atlassian / GitLab / Jira tool integrations to automate developer workflows. Multi-LLM evaluation pipeline benchmarks model quality, cost, and latency; prompts shipped as versioned, deployable artefacts. Treats internal AI as a product with a release process.

Self-Service Multi-Brand Publishing Platform • Burda Forward

- Shared content platform across 200+ publishers and multiple brands with Vue / Angular front-ends and Node.js / TypeScript services; reusable UI component library in Storybook, with Playwright / Cypress / supertest testing in CI. Designed for reusability with zero duplication of publishing logic. Hired the team, set the architecture, and shipped the system that the broader group then operated.

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Seven One Entertainment Group (ProSieben Media SE), Munich

Jun 2024 – Present

Live TV AdTech — backend, full-stack, platform, and AI engineering for 25M+ daily users

- AI: shipped the OpenSearch natural-language interface (Next.js / TypeScript + Claude / Anthropic API) and the GenAI developer agent (see Featured Work) — production LLM applications now used by product managers and engineers.
- Platform: led an end-to-end AWS migration to isolated per-team accounts (Terraform IaC), migrated container workloads from ECS EC2 to Fargate, and integrated AWS Inspector — reducing platform vulnerabilities by 90% and delivering €8,000/month in infrastructure savings.

- Product engineering: designed the real-time event-stream pipeline (AWS Lambda / SQS / SNS / DynamoDB) for dynamic ad insertion, directly powering live business KPIs at 25M+ user scale.
- Developer experience: rebuilt CI/CD with GitLab CI (25 → 8 min, 68% faster; 282 commits & 150 runs/day) and deployed Datadog as the organisation-wide observability platform across all AWS accounts.

Senior Software Engineer / Architect | Burda Forward GmbH (Hubert Burda Media), Munich Oct 2018 – May 2024
Multi-brand publishing group — 6 engineering teams, 200+ publishers

- Product & front-end: architected the multi-brand content platform with Vue and Angular front-ends (TypeScript), designed the CMS rendering shared content across brands, and led international publishing infrastructure for FOCUS Online.
- Engineering: refactored a legacy Java publishing application (150+ German publishers) into a cloud-native Node.js / TypeScript service; built a reusable UI component library in Storybook with Playwright / Cypress end-to-end and supertest API testing in CI — hired, mentored, and grew the team alongside the rewrite.
- Operations: built and ran a Proxy API serving 5M+ daily targeted ad requests with end-to-end ownership of reliability and observability.

Master Thesis Intern (Machine Learning) | BMW Group, Munich Sep 2016 – May 2017
 - Predictive maintenance research using RNNs on BMW battery time-series datasets; classification of BMW makes and models using CNNs.

Data Analyst Intern | Airbus Group Innovations, Munich Feb 2016 – Jul 2016
 - Image processing and virtual flight simulations for flight-health-management systems at Airbus Defence & Space, Manching.

Programmer Analyst | Cognizant Technology Solutions, Kolkata Dec 2011 – Aug 2014
 - Java & Angular enterprise applications for Delta Airlines, Travelers Insurance, and PHH Mortgage.

EDUCATION & CERTIFICATIONS

M.Sc. Computer Science — Technical University of Munich (TUM) 2015 – 2018
Specialisation: Artificial Intelligence & Distributed Systems · Thesis: Predictive Maintenance of Batteries (RNNs), BMW Group

B.Tech. Information Technology — West Bengal University of Technology 2007 – 2011

AWS Certified Solutions Architect – Professional 2026

LANGUAGES

English (C2) • German (C1) • Hindi (Advanced) • Bengali (Native)